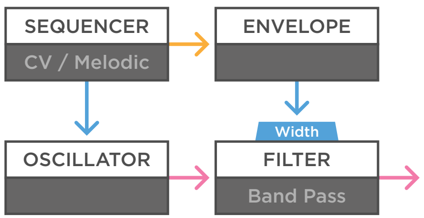
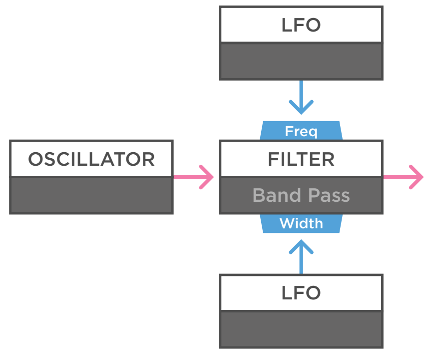
Triple Bandpass Filter

Single Filter Patches

- Audio
- Control Voltage
- Trigger / Gate



(a)



(b)

a - You can use a single bandpass filter to create a very dynamic setup. Just use an oscillator, sequencer and envelope to create a simple voice. In this setup you can use the bandpass filter in different functions. If you set the frequency to minimum, and use the envelope to widen the bandwidth, you create a simple low pas filter setup. But by changing the central frequency, bandwidth, as well as the bipolar attenuators for the incoming envelope, you can morph the function of the filter from low, through band to high pass.

b - Because the filters offer control voltage inputs for both the frequency, as well as the bandwidth, you can create very dynamic modulation. When used as a simple drone voice, for example by feeding the filter a complex waveform, you can use two independent slow LFOs to create movement. When used with enough spread, the filter will move from low to band to high pass, with varying settings.

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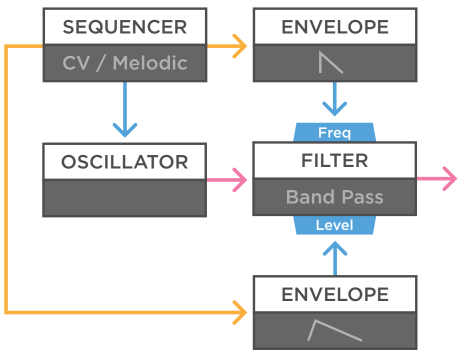
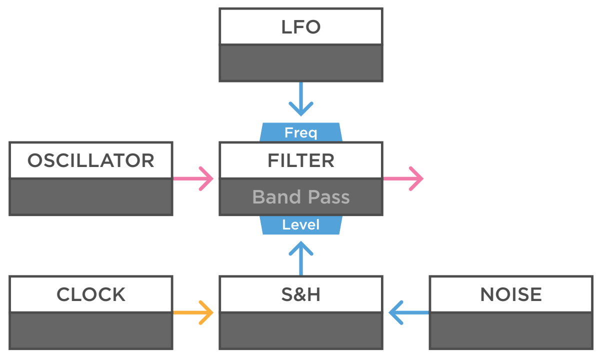


(a)



(b)

a - The ADDAC 603 offers additional functionality because it has cv control over the level output. With moderate input gain, this can be used as a fairly clean VCA, so no external VCA is needed. For example in a simple voice with a sequencer, triggering two envelopes with different settings.

b - The level control is also great with higher gain and feedback settings, effectively becoming a voltage controlled distortion. For example, In this drone setup, with a slow random voltage to the frequency, you can use something like a sample and hold signal to create clocked overdrive.

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MONOTRAIL
TECH TALK

Audio
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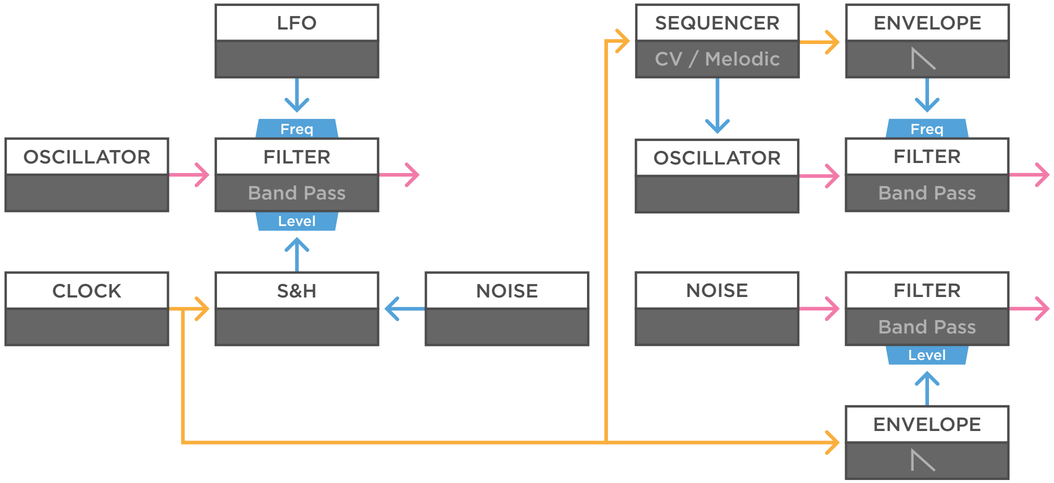


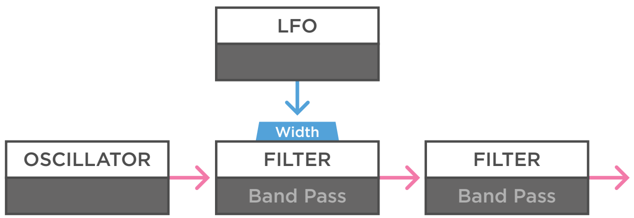
(a)

a - Because the ADDAC 603 offers three independent filters, you can use these setups on individual sounds. This is one example. The voice on the left is similar to the last patch. A second filter is used with a sequencer to create a higher melodic voice. The sequencer is clocked from the first voice. And finally, white noise is sent through the third filter to create a high-hat sound. The output level is modulated with another envelope, clocked with the same central clock.

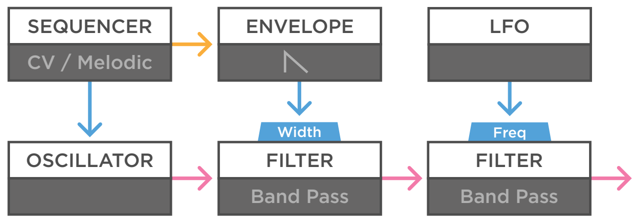
Triple Bandpass Filter

Dual Filter Patches

Audio
Control Voltage
Trigger / Gate



(a)



(b)

a - This is a simple setup, with two filters used in series. With each bandpass filter being so flexible, you can use this setup for many different functions. For example, you can use the first filter like a ow pass filter, and the second like a high pass filter. The first to shape the high end, and the second to cut out low frequencies, and create controlled mid-range drone textures. A simple LFO is modulating the first filter in this patch.

b - But in a dual setup, bandpass filters can add unique textures to your sound. For example in a simple voice, where the first filter is used in a low pass setup, opened by an envelope. Then the sound goes through the second filter, used as a narrow band pass with some resonance. You can use a slow source like an LFO or random voltage to create subtle dynamics.

Triple Bandpass Filter
Dual Filter Patches

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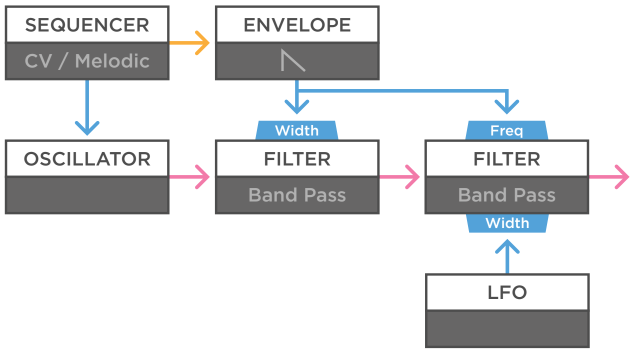
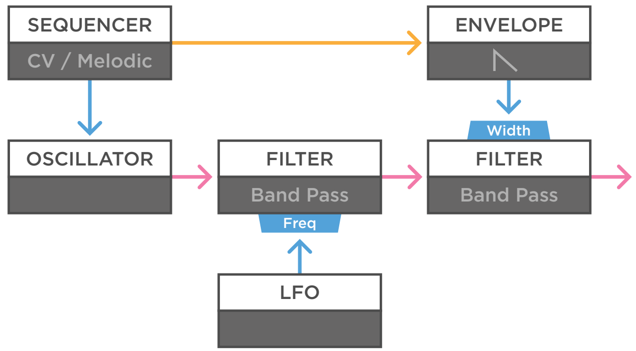


(a)



(b)

a - In a similar setup, you can experiment by sending a copy of the envelope to both filters at the same time. You can use the envelopes to modulate either the frequency or bandwidth, And of course add LFOs wherever you like, to create more motion over time.

b - You can create different textures in the same setup, by changing how the filters are used. For example, you can use the first filter as a narrow band pass with a lot of feedback. And have a slow LFO sweep that filter up and down. Then use a triggered envelope to open the width of the second filter, used more as a low pass, to tame the sound again.

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Dual Filter Patches

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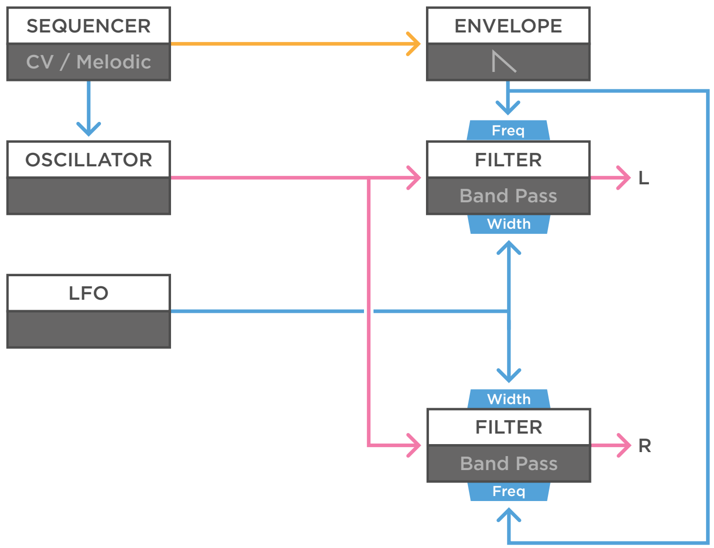
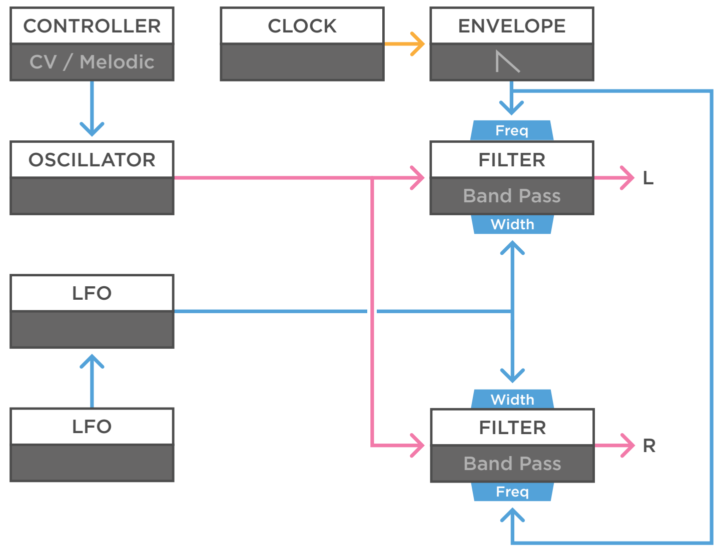
MONOTRAIL
TECH TALK



(a)



(b)

a - Using two filters in parallel creates different options. For example, you can easily create wonderful stereo sounds. Set up a simple voice and run it through two filters with the same settings. Here the envelopes are used on the frequency of both filters. Then, just use a single LFO to modulate the band width of both filters. Make sure to use the attenuverters to create an inverted effect on one of them. This is what creates subtle shifts in the stereo field.

b - Here is a very similar setup, but this time a controller is used to send a 1v/oct signal to the oscillator. The envelope opening both filters is triggered by a free running clock, and an addition LFO is used to modulate the speed of the LFO that's causing the stereo width.

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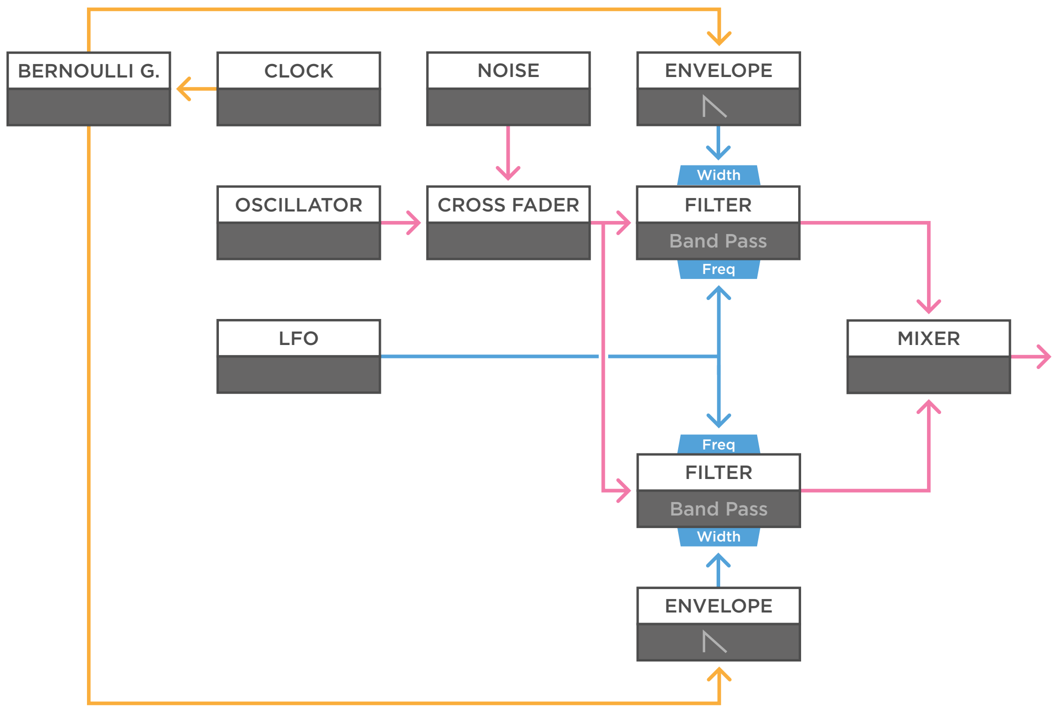

(a)

a - You don't need to use parallel filters in stereo. For example, you can run the same sound through two bandpass filters, and then mix the result together. Here a cross fader is used to mix an oscillator with white noise. When used with more narrow bands, and audible resonance peaks, you can create very dynamic sounds.

In this example an LFO is modulating the frequency of both filters, one with inversed polarity. And finally a Bernoulli gate is used to divide a steady clock over two envelopes, modulating the band width of the filters.

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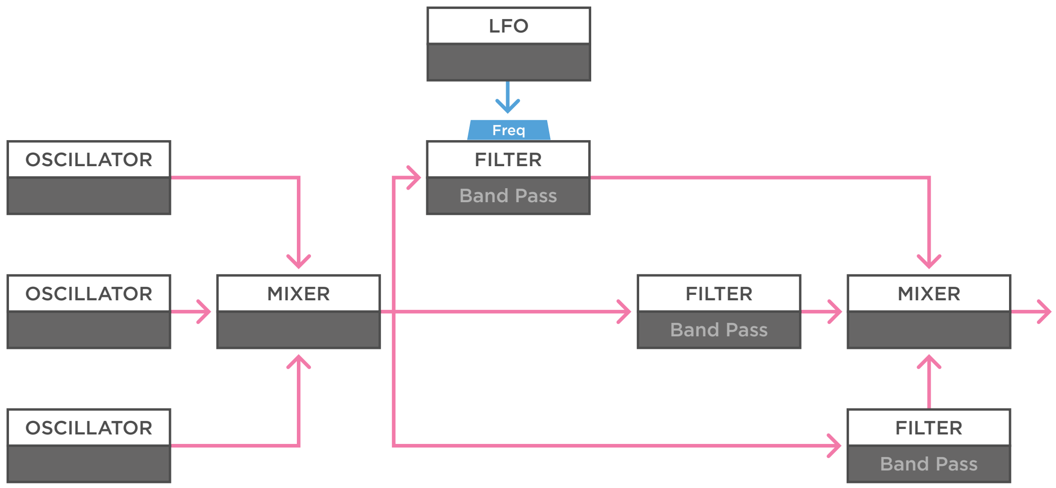

(a)

a - A setup with all three filters of the ADDAC 603 can be used for powerful and detailed sound sculpting. To give you an idea, feed a mix of steady saw wave oscillators into the three band pass filters with varying setting.

The flowchart shows a mixer at the end for clarity, but you can use the mix output of the ADDAC 603. Just a single LFO is modulating the frequency of one of the filters to create a bit of automated motion. But you can use the controls to sculpt the sound manually.

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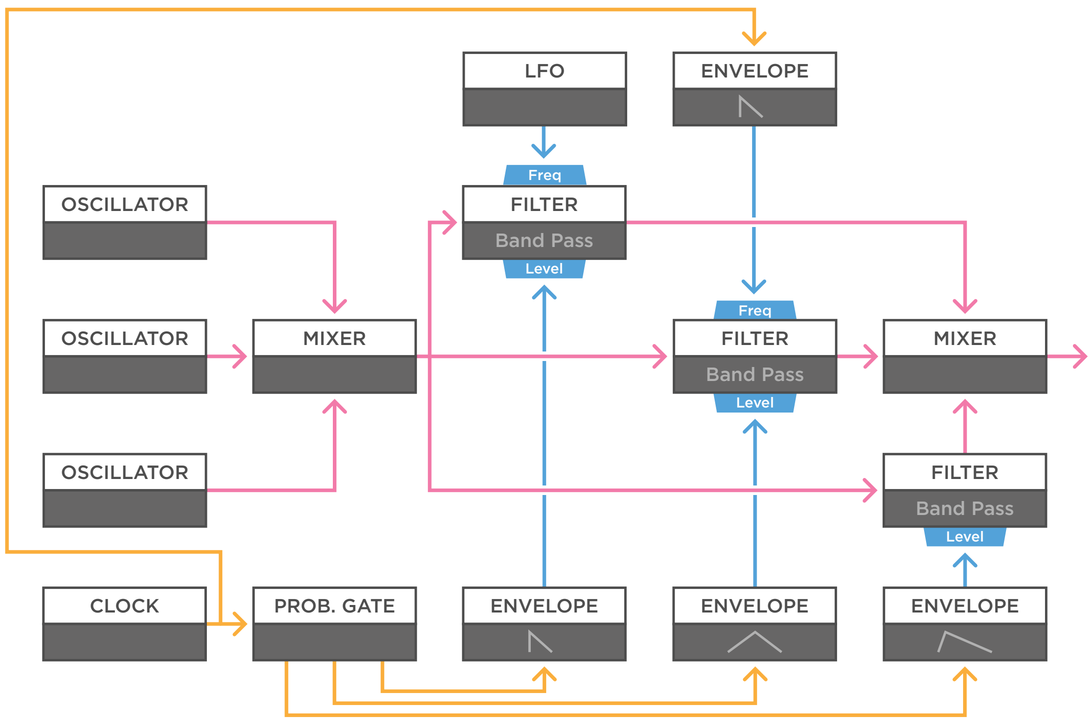

(a)

a - To turn the previous patch into a self-playing setup, you can add time based motion. In this patch, a clock is driving a probabilistic gate, sending the pulse to either of three envelopes, each with different settings. Those envelopes control the level of each of the filters. Finally, the clock also drives a short plucky envelope, opening the second filter, used with higher resonance.

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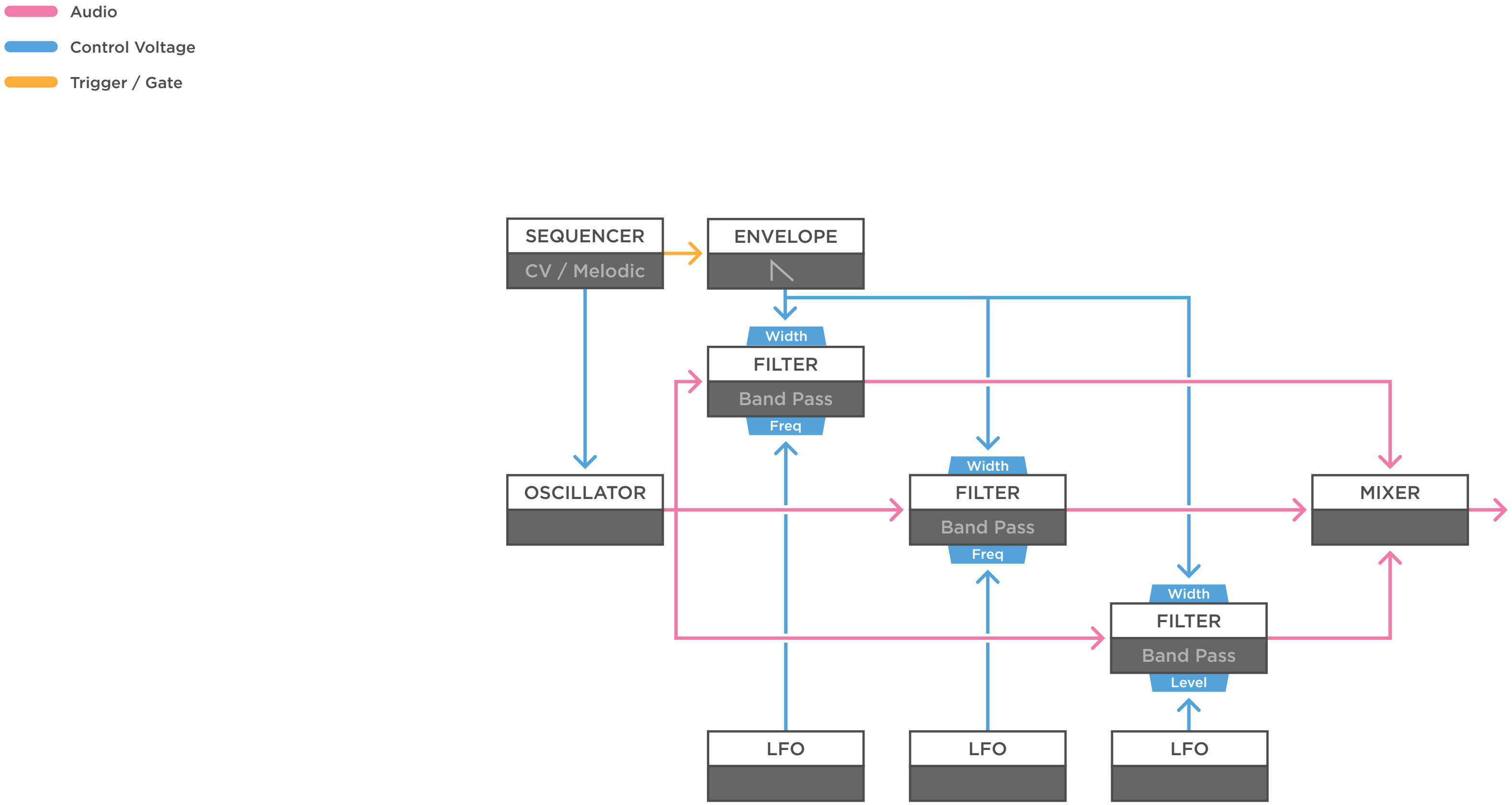

(a)

a - Here is an example in a more melodic setup. A single oscillator is sent into al three filters, and mixed together, using the mix output. A sequencer is used to tune the oscillator, and trigger a single plucky envelope, that is modulating the bandwidth of all three filters.

Finally, three independent free running LFOs are used. Two are modulating the frequency, one is modulating the level. With each filter offering a unique result, you can use the level control of each filter to make a mix, and create complex layered sounds.

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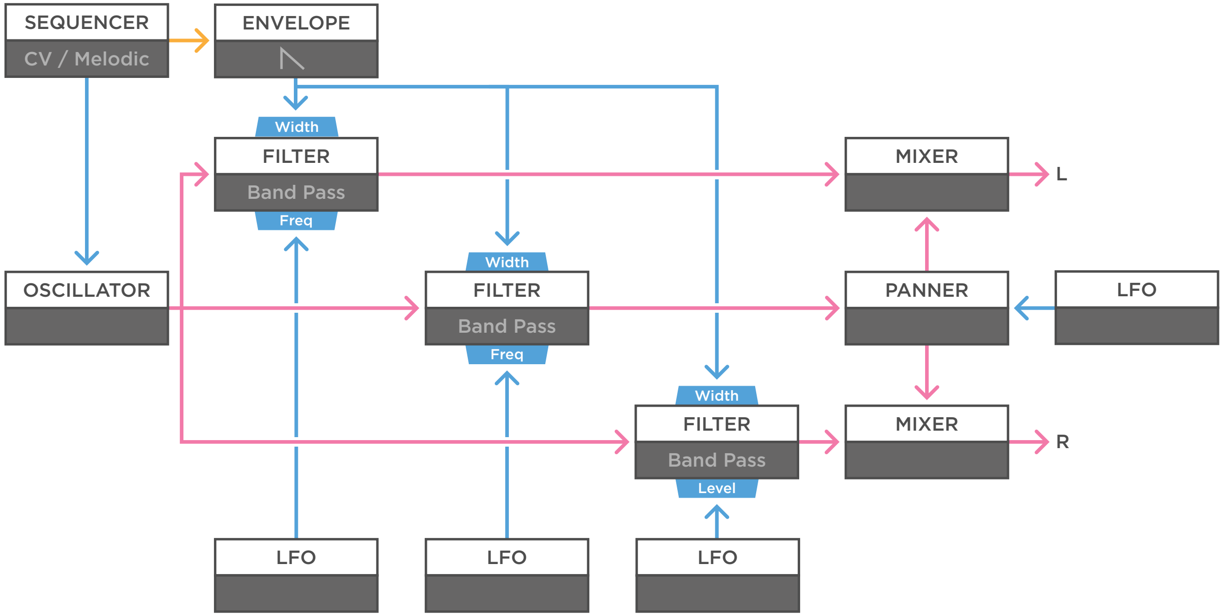

(a)

a - You can still use the individual outputs of the filters to create other effects, like complex stereo setups. Here, the same setup as the last patch is used, but instead of using the sum output, two filters are sent to individual mixers, to creates a stereo signal. Then the third filter, is sent to a panner, and those left and right outputs are mixed in with the other stereo signal. Another LFO is used to modulate the panner.